



SURGR
VERIFIED OUT LOUD

Every verbal order in the OR, verified out loud.

A real-time closed-loop safety copilot for the operating room. It listens, checks every drug order against its read-back, tracks the WHO checklist, speaks up when a loop does not close, and writes the audit record.

ASSEMBLYAI VOICE AGENT HACKATHON · SEPTEMBER 2026 · [GITHUB.COM/MRNETWORK0001/SURGR](https://github.com/MRNETWORK0001/SURGR)

The loop breaks quietly

Drugs in surgery are ordered by voice. The standard is closed-loop communication: repeat the order back, get it confirmed, then act. Aviation made that mandatory decades ago. In an operating room it depends on people remembering under noise, fatigue and time pressure. Nothing beeps when a read-back does not happen.

1 in 20

perioperative
medication
administrations involve
an error or adverse drug
event

Nanji et al., Anesthesiology, 2016

11% → 7%

major complications
after the WHO Surgical
Safety Checklist; deaths
1.5% → 0.8%

Haynes et al., NEJM, 2009

~2 in 3

sentinel events involve a
breakdown in team
communication

The Joint Commission

310M+

major operations
performed worldwide
every year

Weiser et al., 2016

Listen, understand, intervene, document

01

Listen

Room audio streams to AssemblyAI. Medical mode gets drug names and doses right; speaker labels tell surgeon, anesthesiologist and nurse apart on one microphone.

02

Understand

A deterministic classifier parses orders, read-backs and checklist statements in under a millisecond. Ambiguous drug talk goes to the LLM Gateway under a strict JSON schema.

03

Intervene

Every order gets ten seconds to be read back. Wrong dose, wrong drug, wrong route or silence, and Surgr speaks the alert into the room through the Voice Agent.

04

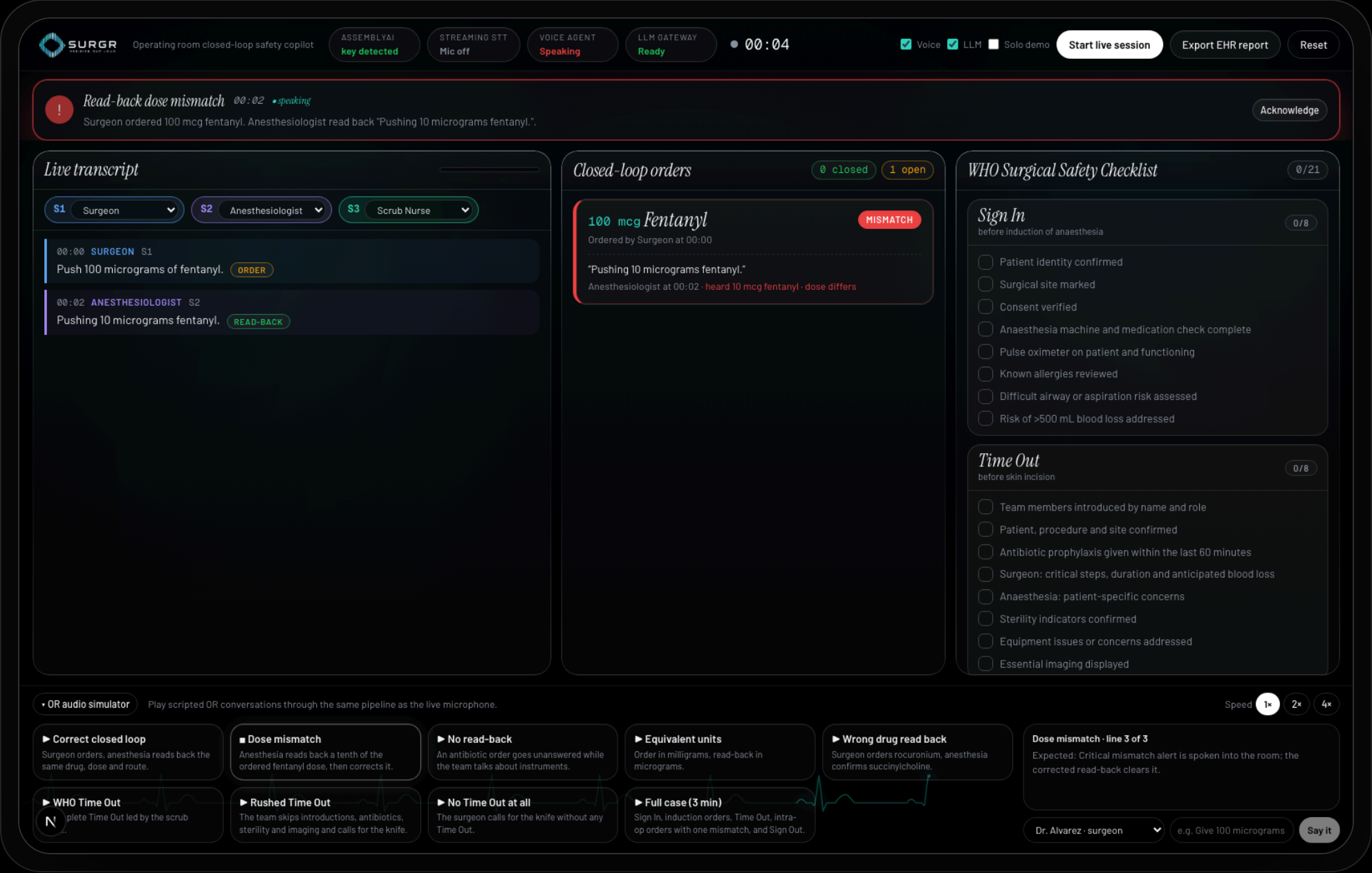
Document

An operative communication record where every order, read-back, checklist item and safety event links to the exact second in the audio. PDF, Telegram, email, spoken debrief.

“Safety alert. Read-back dose mismatch. The surgeon ordered one hundred micrograms fentanyl. Anesthesia read back ten micrograms. Please stop and re-confirm.”

What the room hears about one second after a wrong read-back. This sentence is the product.

THE COCKPIT



One screen for the whole room. Nobody has to look at it.

- Transcript by role, every order and read-back tagged as it lands.
- Order board with a ten-second countdown; a mismatch goes red and stays red until a correct read-back arrives.
- WHO checklist that ticks itself from speech; a phase that ends with gaps, or an incision without a Time Out, raises a spoken alert.
- Simulator with nine scripted scenarios so any judge can trigger a wrong dose without an operating room.

The room can ask it questions

The Voice Agent listens to the room for the whole case. Anything not addressed to Surgr is answered with silence, discarded before it plays. Say its name and it answers from live state through JSON-schema tools.

`get_open_orders``get_last_order``get_checklist_status``get_session_summary``get_recent_alerts``acknowledge_alerts`

“Surgr, what is still open?” → “You have one open order for two grams of cefazolin.”

Verified against the live Voice Agent API: tool call issued, result returned, answer spoken.

Pick a scenario, or type a line as any team member.

Ask Surgr ANSWERED

×

“Sergio, what was the last order?”

The last order was two grams of cefazolin, which has been confirmed.

Dr. Alvarez · surgeon



(e.g. Give 100 micrograms)

Say it

Surgr Operative Communication Record

Generated 9/27/2026, 10:37:43 PM · narrative by qwen3.5-4b-32k-fast via AssemblyAI LLM Gateway · every entry links to the audio timestamp

Download JSON

Print

Close

Deliver

Download PDF

Spoken debrief

Send to Telegram

name@hospital.org

Email PDF

00:47
SESSION

4/4
ORDERS CLOSED

100%
CLOSED-LOOP RATE

1.1s
MEAN READ-BACK TIME

1
SAFETY EVENTS

PROCEDURE SUMMARY

A right knee arthroscopy with partial meniscectomy was performed on John Doe following a complete sign-in and time-out. The team confirmed patient identity, site marking, antibiotic prophylaxis, and critical steps before induction. The procedure concluded with correct sponge, needle, and instrument counts, and the specimen was properly labeled.

MEDICATIONS (CLOSED-LOOP RECORD)

DRUG	DOSE	ROUTE	ORDERED	READ-BACK	STATUS	NOTE
propofol	200 mg	not stated	Surgeon · 00:02	Anesthesiologist · 00:03	CONFIRMED	Propofol 200 mg ordered by surgeon; read-back confirmed immediately by anesthesiologist.
fentanyl	100 mcg	not stated	Surgeon · 00:04	Anesthesiologist · 00:04	CONFIRMED	Fentanyl 100 mcg ordered by surgeon; read-back confirmed immediately by anesthesiologist.
ketorolac	30 mg	IV	Surgeon · 00:13	Anesthesiologist · 00:14	CONFIRMED	Ketorolac 30 mg IV ordered by surgeon; read-back confirmed immediately by anesthesiologist.
ondansetron	4 mg	not stated	Surgeon · 00:15	Anesthesiologist · 00:17	CONFIRMED	Ondansetron 4 mg ordered by surgeon; initial read-back error (8 mg) corrected by anesthesiologist before administration.

WHO CHECKLIST COMPLIANCE

Sign In

COMPLETE

✓ Patient identity confirmed 00:00

✓ Surgical site marked 00:00

✓ Consent verified 00:00

✓ Anaesthesia machine and medication check complete 00:00

✓ Pulse oximeter on patient and functioning 00:00

Time Out

COMPLETE

✓ Team members introduced by name and role 00:06

✓ Patient, procedure and site confirmed 00:08

✓ Antibiotic prophylaxis given within the last 60 minutes 00:09

✓ Surgeon: critical steps, duration and anticipated blood loss 00:10

Sign Out

COMPLETE

✓ Name of the procedure recorded 00:18

✓ Instrument, sponge and needle counts correct 00:19

✓ Specimen labelled 00:20

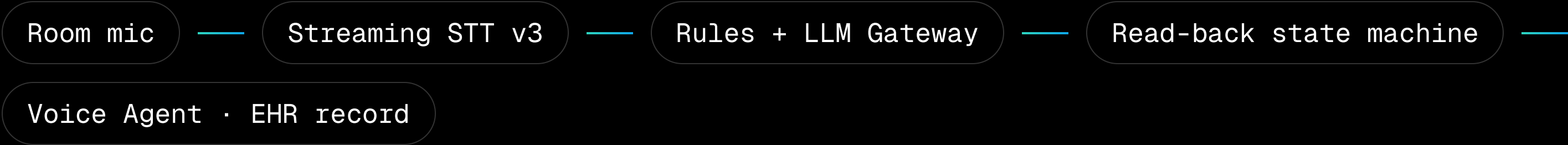
✓ Equipment problems identified 00:20

✓ Key concerns for recovery and management 00:21

The audit writes itself

- Closed-loop rate, mean read-back time, every medication with order and read-back timestamps.
- WHO compliance per phase, safety events with how they were resolved, recommendations.
- Facts and timestamps always come from Surgr's own state; the LLM only writes narrative, and the record says so.
- Delivered as PDF, to Telegram, by email, through the device share sheet, or read aloud as a spoken debrief.

Four products, one safety loop



LAYER	ASSEMBLYAI PRODUCT	HOW SURGR USES IT
Live transcription	Streaming v3 · universal-3-6-pro	Medical mode formats drugs and doses; 47-drug keyterm list; clinician silence thresholds; short-lived browser tokens.
Who is speaking	speaker_labels	True diarization on one microphone, so a read-back from the person who gave the order is caught.
Understanding	LLM Gateway	JSON-schema classification of ambiguous drug talk; report narrative. Budget read from rate-limit headers, never blocks the safety loop.
Voice in the room	Voice Agent API	Alerts via reply.create; continuous room listening with silent-reply suppression; six client tools answer questions from live state.

Every integration tested against the live services

Streaming

A 78-second three-voice rehearsal streamed to AssemblyAI transcribed all nine lines verbatim with two speaker labels and doses written as digits by medical mode.

Voice Agent

A mismatch alert was spoken word for word (eleven seconds of audio). Tool calling confirmed: question in, **get_open_orders** called, answer spoken.

Live microphone

A real session in Chrome: two voices separated, three alerts spoken at the right moments, orders closed, report exported, printed and delivered.

<1ms

rule classification per utterance

≈1s

from a wrong read-back to the spoken alert

10s

read-back window before a timeout alert

Nine scripted scenarios run through the engine as a regression suite. Solo demo mode and a rehearsal audio file let anyone test without a second person.

Built for the people administering the drug

Anesthesiologist

A second check on every dose before the syringe goes in. The person who benefits most, because they are the one administering.

Surgeon

Confirmation that what was said is what is being given, without policing it. Orders, read-backs and alerts documented for them.

Nurses

The checklist tracks itself. No clipboard, no chasing eight items while the room moves on.

Quality & safety office

Objective closed-loop and checklist compliance per room and per team, and a defensible timestamped record when an incident is reviewed.

A day with it

A microphone, a speaker, Surgr on the wall or nowhere visible. The team works as before. Once or twice a week it says “read-back mismatch, please re-confirm”. That sentence is the product.

Priced per operating room

TAM

310M+

major operations a year worldwide, in an estimated 250,000+ operating rooms across hospitals and surgery centres.

SAM · UNITED STATES FIRST

≈\$240M/yr

≈40,000 US operating rooms × \$500 per OR per month. Sold to hospital quality and perioperative leadership.

BEACHHEAD

≈\$10M/yr

≈1,000 simulation centres and residency programmes at \$10k per site licence, where instant spoken feedback needs no regulatory clearance.

Revenue streams

Per-OR subscription for live use · site licences for simulation and training · analytics tier for quality offices · integration services for EHR export. Figures are planning estimates, not audited market data.

Others document or review. Surgr intervenes.

CATEGORY	EXAMPLES	WHAT THEY DO	WHAT SURGR DOES DIFFERENTLY
Ambient clinical scribes	Abridge, Nuance DAX, Ambience	Turn conversations into notes after the fact.	Acts during the case, in the room, out loud.
OR black-box analytics	Surgical Safety Technologies	Record the OR for retrospective review.	Closes the loop in the moment; the record is a by-product.
Checklist apps	Digital WHO checklists	Someone taps items on a tablet.	Items tick from speech; skipped phases are spoken alerts.
Hackathon look-alikes	OR and resuscitation voice recorders	Same instinct: transcribe and flag.	Deterministic dose and unit logic, speaker-aware self-read-back detection, a room that can ask questions, delivery to Telegram and a spoken debrief, all verified live.

Moat: the deterministic safety engine and the evidence base it produces, not the transcription layer.

Three stages into the operating room

STAGE 1 · NOW

Simulation and training

Residency programmes and simulation labs. Instant spoken feedback teaches closed-loop discipline. No regulatory barrier.

STAGE 2

Silent mode in real ORs

Compliance analytics per room and per team, no intervention. Builds the evidence base and the pilot relationships.

STAGE 3

Live intervention

After clinical validation and regulatory clearance. Multilingual rooms, EHR integration, on-premise audio.

What we say plainly

Surgr is a prototype, not a medical device. Similar voices can share a speaker label. Hackathon accounts get two LLM Gateway requests a minute, so the deterministic engine does the real-time work and the record says when the narrative fell back to the local summary.



SURGR
VERIFIED OUTLOUD

Aviation cockpits have mandatory read-backs and a voice recorder. Operating rooms have neither.

Surgr gives them both.

LIVE APP · SEE SUBMISSION LINK · [GITHUB.COM/MRNETWORK0001/SURGR](https://github.com/MRNETWORK0001/SURGR) · BUILT ON ASSEMBLYAI